

Python_Basics

1. What is Programming?

Programming means writing instructions for a computer to perform a task.

Example: Example: `print("Hello World")`

2. Python & Uses:

Python is a simple, high-level programming language used for many types of applications.

Example: Example: Python is used for AI, data science, web development, and automation.

3. Python Syntax:

Syntax means the rules we follow to write correct Python code.

Example: Example: `print("Hello")`

4. Python Virtual Environment:

A virtual environment creates a separate space for a project's Python packages.

Example: Example: `python -m venv myenv`.

5. Python Operators:

Operators are symbols used to perform calculations and comparisons.

Example: Example: `10 + 5` gives 15; `10 > 5` gives True.

6. Python Data Types:

Data types tell Python what kind of value a variable contains.

Example: Example: `name = "Divya"` (string), `age = 36` (integer), `active = True` (boolean).

7. Lists, Sets, Tuples, Dictionary:

These are Python collections used to store multiple values.

Example: Example: `list = [10, 20, 30]`; `tuple = (10, 20)`; `set = {1, 2, 3}`; `dictionary = {'name': 'Divya'}`.

8. Python Control Flow:

Control flow decides which part of the program should run based on conditions.

Example: Example: `if age >= 18: print('Adult')`

9. Nested If Statements:

A nested if is an if statement placed inside another if statement.

Example: Example: if age >= 18: then check another condition such as if has_id == True.

10. Python Loops:

Loops repeat a block of code multiple times.

Example: Example: for i in range(3): print(i) → prints 0, 1, 2.

11. Python Function:

A function is a reusable block of code that performs a specific task.

Example: Example: def add(a, b): return a + b.

12. Python Modules:

A module is a Python file containing reusable code such as functions and classes.

Example: Example: import math; math.sqrt(25) gives 5.

13. Error Handling:

Error handling lets a program handle errors without stopping unexpectedly.

Example:Example: try: result = 10 / 0 except ZeroDivisionError: print('Cannot divide by zero')

14. Python File Handling:

File handling is used to create, read, write, and update files.

Example: Example: with open('data.txt', 'r') as file: data = file.read()